

```
In [17]: !git clone https://github.com/ChristosPaterakisntua/digit-recogniser-mnist
fatal: destination path 'digit-recogniser-mnist' already exists and is not
an empty directory.
```

```
In [18]: !git -C /content/digit-recogniser-mnist pull
Already up to date.
```

```
In [19]: import os
print(os.listdir("/content/digit-recogniser-mnist"))

['predict.py', 'test.py', '__pycache__', 'utils.py', 'dataset.py', '.gitignore', '.git', 'evaluate.py', 'train.py', 'loader.py', 'model.py']
```

```
In [20]: import sys
sys.path.insert(0, "/content/digit-recogniser-mnist")
```

```
In [33]: from __future__ import annotations

from dataset import load_mnist
from model import CNNDigitRecogniserModel, MLPDigitRecogniserModel
from train import train, CHECKPOINT_PATH
from evaluate import load_trained_model, evaluate
from loader import load_raw_image
from predict import predict
from utils import select_device, ask_yes_or_no
```

```
In [31]: device = select_device()
print(device)

train_model: bool = ask_yes_or_no("Train a new model?")
if train_model:
    train_cnn: bool = ask_yes_or_no("Train the cnn model?")
    train_mlp: bool = ask_yes_or_no("Train the mlp model?")
```

cuda

```
In [ ]: if train_cnn:
    train_dataloader = load_mnist()
    model = CNNDigitRecogniserModel()
    train(
        model=model,
        data_loader=train_dataloader,
        device=device,
        model_id="CNN_1",
        checkpoint_path=CHECKPOINT_PATH,
    )

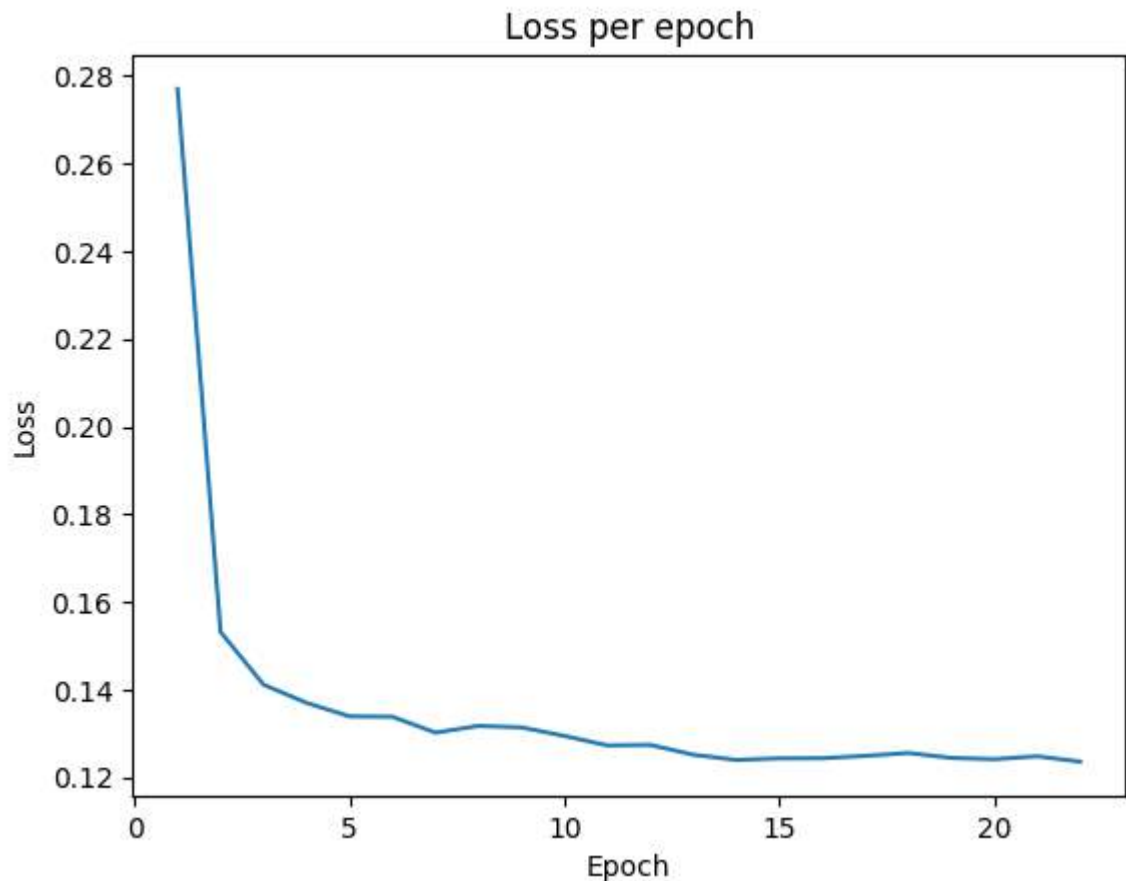
else:
    path = input("Path to trained model: ")
    model = load_trained_model(
        path=path,
        device=device,
    )
```

Epoch 22/50

Training loss: 0.12349814825853477

Checkpoint saved: saved_models/checkpoint.pt

Early stopping at epoch: 22



Final model saved: saved_models/model_CNN_1.pt

```
In [24]: evaluation = evaluate(
          model=model,
          device=device,
        )
        print(f"Evaluation: {100 * evaluation:.2f}%")
```

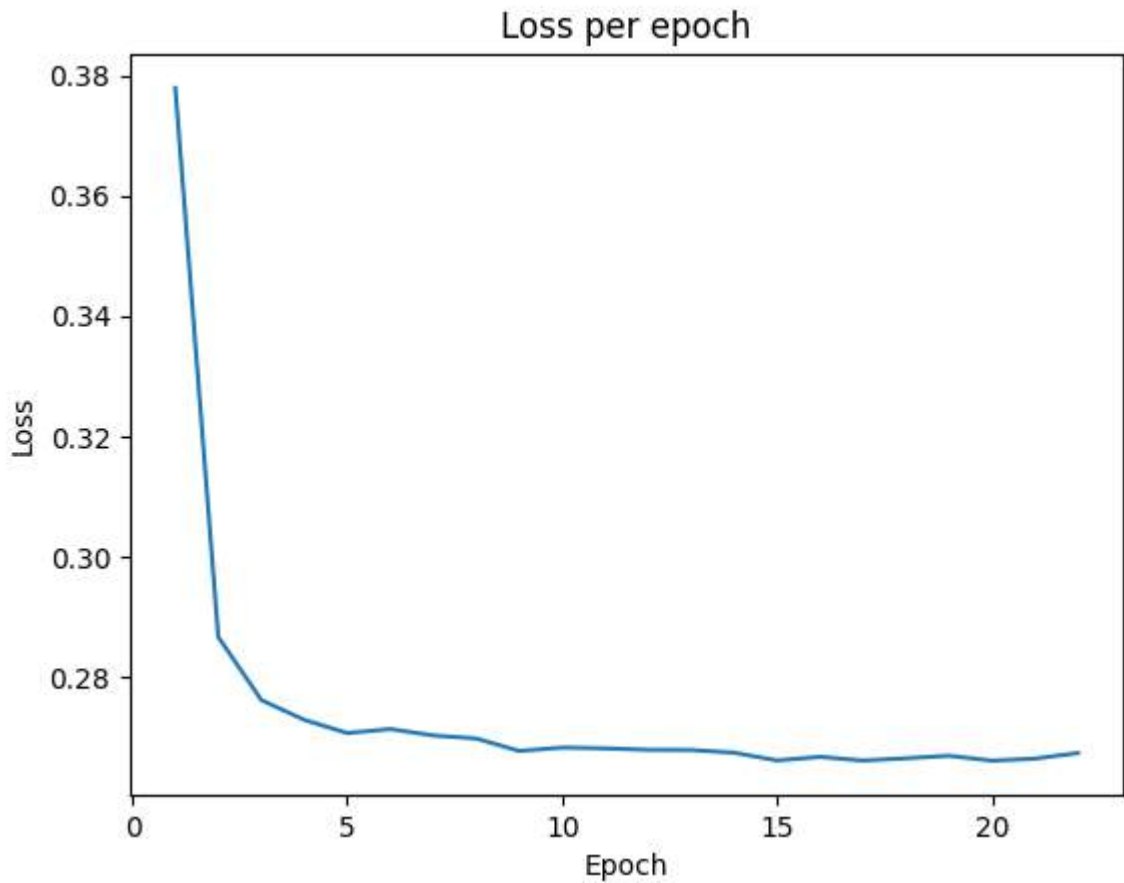
Evaluation: 96.58%

```
In [34]: if train_mlp:
          train_dataloader = load_mnist()
          model1 = MLPDigitRecogniserModel()
          train(
              model=model1,
              data_loader=train_dataloader,
              device=device,
              model_id="MLP_1",
              checkpoint_path=CHECKPOINT_PATH,
          )

      else:
          path = input("Path to trained model: ")
          model1 = load_trained_model(
              path=path,
              device=device,
          )
```

Epoch 1/50
Training loss: 0.377881414491528
Checkpoint saved: saved_models/checkpoint.pt
Epoch 2/50
Training loss: 0.2866129800414395
Checkpoint saved: saved_models/checkpoint.pt
Epoch 3/50
Training loss: 0.27613423488265837
Checkpoint saved: saved_models/checkpoint.pt
Epoch 4/50
Training loss: 0.2728397174869897
Checkpoint saved: saved_models/checkpoint.pt
Epoch 5/50
Training loss: 0.2706439980765572
Checkpoint saved: saved_models/checkpoint.pt
Epoch 6/50
Training loss: 0.27130315735936167
Checkpoint saved: saved_models/checkpoint.pt
Epoch 7/50
Training loss: 0.2702470910367168
Checkpoint saved: saved_models/checkpoint.pt
Epoch 8/50
Training loss: 0.2697410562068884
Checkpoint saved: saved_models/checkpoint.pt
Epoch 9/50
Training loss: 0.2676693604203465
Checkpoint saved: saved_models/checkpoint.pt
Epoch 10/50
Training loss: 0.2682276183801587
Checkpoint saved: saved_models/checkpoint.pt
Epoch 11/50
Training loss: 0.2680978704184915
Checkpoint saved: saved_models/checkpoint.pt
Epoch 12/50
Training loss: 0.2678548556712534
Checkpoint saved: saved_models/checkpoint.pt
Epoch 13/50
Training loss: 0.2678254833109095
Checkpoint saved: saved_models/checkpoint.pt
Epoch 14/50
Training loss: 0.26737875856210785
Checkpoint saved: saved_models/checkpoint.pt
Epoch 15/50
Training loss: 0.2660898775450575
Checkpoint saved: saved_models/checkpoint.pt
Epoch 16/50
Training loss: 0.26668558015616145
Checkpoint saved: saved_models/checkpoint.pt
Epoch 17/50
Training loss: 0.26606954337210237
Checkpoint saved: saved_models/checkpoint.pt
Epoch 18/50
Training loss: 0.266454612994683
Checkpoint saved: saved_models/checkpoint.pt
Epoch 19/50
Training loss: 0.2668688920833714
Checkpoint saved: saved_models/checkpoint.pt
Epoch 20/50
Training loss: 0.2660516025829323
Checkpoint saved: saved_models/checkpoint.pt

Epoch 21/50
Training loss: 0.26639519589056726
Checkpoint saved: saved_models/checkpoint.pt
Epoch 22/50
Training loss: 0.2673193224204006
Checkpoint saved: saved_models/checkpoint.pt
Early stopping at epoch: 22



Final model saved: saved_models/model_MLP_1.pt

```
In [35]: evaluation = evaluate(  
        model=model1,  
        device=device,  
    )  
    print(f"Evaluation: {100 * evaluation:.2f}%")
```

Evaluation: 92.20%

```
In [ ]: do_prediction: bool = ask_yes_or_no("Want a prediction?")  
  
if do_prediction:  
    print("Type exit to stop")  
    while True:  
        path = input("Path: ").strip()  
        if path.lower() == "exit":  
            break  
        img_tensor = load_raw_image(path)  
        prediction = predict(  
            model=(model if model is not None else model1),  
            image_tensor=img_tensor,  
            device=device,  
        )  
        print(f"Model predicted: {prediction}")
```